

Tutorial 3: Integer Programming

Modeling, Branch-and-Bound, and Logical Constraints

IEDA 1250

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Tutorial overview

In this tutorial, we will study:

- ① why integer programming is needed;
- ② how to build an assignment model;
- ③ how LP relaxation is used in Branch-and-Bound;
- ④ how to model logical constraints with binary variables;
- ⑤ how to formulate a fixed-charge facility opening problem.

Main idea

Integer programming is useful when decisions are discrete: assign or not assign, open or not open, choose or not choose.

Why integer programming?

Suppose a company has 3 candidates and 3 jobs.

Number of possible assignments = $3! = 6$.

This can be solved by enumeration.

If the company has 100 candidates and 100 jobs:

Number of possible assignments = $100!$.

Motivation

Enumeration becomes impossible very quickly. We need optimization models and algorithms.

Binary decision variables

A binary variable takes value 0 or 1:

$$x_i \in \{0, 1\}.$$

It usually represents a yes-or-no decision:

$$x_i = \begin{cases} 1, & \text{if option } i \text{ is selected,} \\ 0, & \text{otherwise.} \end{cases}$$

Examples

Activate a project, open a warehouse, assign a person to a job, or choose an arc in a route.

Assignment problem

Let:

$$x_{ij} = \begin{cases} 1, & \text{if person } i \text{ is assigned to job } j, \\ 0, & \text{otherwise.} \end{cases}$$

If $x_{23} = 1$, then person 2 is assigned to job 3.

If $x_{23} = 0$, then person 2 is not assigned to job 3.

Variable restriction

$$x_{ij} \in \{0, 1\}.$$

Assignment model

Each job gets exactly one person:

$$\sum_i x_{ij} = 1 \quad \text{for each job } j.$$

Each person gets exactly one job:

$$\sum_j x_{ij} = 1 \quad \text{for each person } i.$$

If c_{ij} is the matching score:

$$\max \sum_i \sum_j c_{ij} x_{ij}.$$

$$x_{ij} \in \{0, 1\}.$$

LP relaxation

An integer program may have restrictions such as:

$$x_i \in \{0, 1\} \quad \text{or} \quad x_i \in \mathbb{Z}_+.$$

The LP relaxation ignores the integer requirement:

$$x_i \in \{0, 1\} \quad \Rightarrow \quad 0 \leq x_i \leq 1,$$

$$x_i \in \mathbb{Z}_+ \quad \Rightarrow \quad x_i \geq 0.$$

For maximization

The LP relaxation gives an upper bound because it allows a larger feasible region.

LP relaxation may be fractional

Consider:

$$\begin{array}{ll}\max & 200x_1 + 100x_2 \\ \text{s.t.} & 1.5x_1 + 2x_2 \leq 6, \\ & 3x_1 + x_2 \leq 9, \\ & x_1, x_2 \in \mathbb{Z}_+.\end{array}$$

The LP relaxation may produce:

$$(x_1, x_2) = (2.667, 1).$$

This is not feasible for the original IP.

Key point

LP relaxation is useful for bounds, but fractional solutions are not integer feasible solutions.

Branch-and-Bound

Branch-and-Bound solves IPs by repeatedly using LP relaxations.

- ① Solve the LP relaxation of a node.
- ② If the solution is integer, record it as an incumbent.
- ③ If the solution is fractional, branch on a fractional variable.
- ④ Use bounds to decide whether a node can be ignored.

Vocabulary

UB: optimistic value from LP relaxation.

Incumbent: best integer feasible solution found so far.

B&B example: root relaxation

$$\begin{array}{ll}\max & 40x_1 + 90x_2 \\ \text{s.t.} & 9x_1 + 7x_2 \leq 56, \\ & 7x_1 + 20x_2 \leq 70, \\ & x_1, x_2 \in \mathbb{Z}_+.\end{array}$$

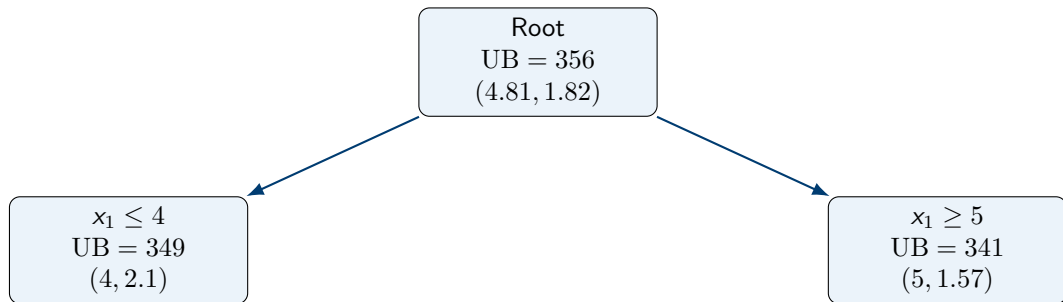
Root LP relaxation:

$$(x_1, x_2) = (4.81, 1.82), \quad \text{UB} = 356.$$

Since $x_1 = 4.81$ is fractional, branch into:

$$x_1 \leq 4 \quad \text{or} \quad x_1 \geq 5.$$

B&B example: first branch



Observation

A smaller upper bound gives lower priority, but it does not automatically justify pruning.

B&B example: finding an incumbent

Explore the node $x_1 \leq 4$:

$$(x_1, x_2) = (4, 2.1), \quad \text{UB} = 349.$$

Branch on $x_2 = 2.1$:

$$x_2 \leq 2 \quad \text{or} \quad x_2 \geq 3.$$

For $x_1 \leq 4$, $x_2 \leq 2$, the LP solution is:

$$(x_1, x_2) = (4, 2).$$

This is integer feasible, with objective:

$$40(4) + 90(2) = 340.$$

B&B example: pruning by bound

The current incumbent is:

$$z = 340.$$

Now consider the sibling node:

$$x_1 \leq 4, \quad x_2 \geq 3.$$

Its LP relaxation gives:

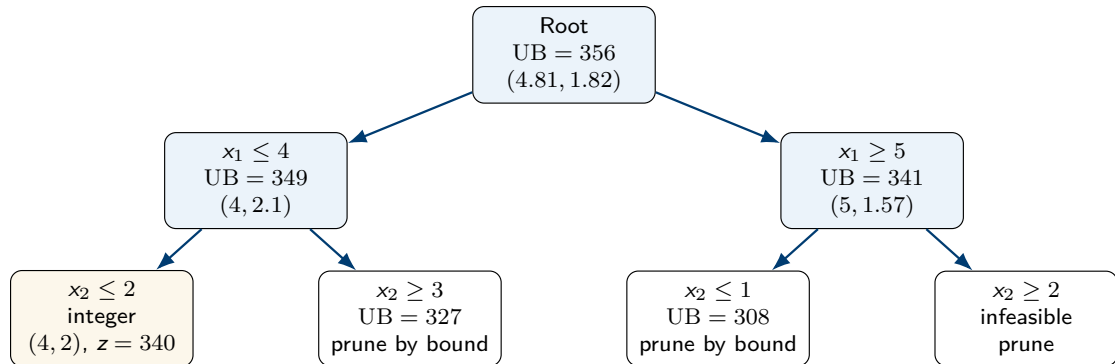
$$(x_1, x_2) = (1.42, 3), \quad \text{UB} = 327.$$

Since this is a maximization problem:

$$\text{best integer value in this node} \leq 327 < 340.$$

Therefore, this node cannot contain the global optimum and can be pruned.

B&B example: complete tree



Final result

All active nodes are closed. The optimal integer solution is $(x_1, x_2) = (4, 2)$, with $z = 340$.

Logical constraints with binary variables

Let $x_i \in \{0, 1\}$.

If option 2 can be selected only if option 1 is selected:

$$x_2 \leq x_1.$$

Options 4 and 5 cannot both be selected:

$$x_4 + x_5 \leq 1.$$

At least one of options 3, 4, and 5 must be selected:

$$x_3 + x_4 + x_5 \geq 1.$$

At most two of options 1, 2, and 3:

$$x_1 + x_2 + x_3 \leq 2.$$

Big-M modeling

Suppose:

$$y = \begin{cases} 1, & \text{if a warehouse is opened,} \\ 0, & \text{otherwise,} \end{cases}$$

and q is the shipment amount.

If shipment is possible only when the warehouse is open:

$$q \leq My.$$

If M is the capacity B , then:

$$q \leq By.$$

Interpretation

If $y = 0$, then $q = 0$. If $y = 1$, then $q \leq B$.

Fixed-charge facility opening problem

A company can open warehouses and ship products to customers.

	Fixed cost	Capacity	C1	C2	C3
W1	400	80	4	6	9
W2	500	100	5	4	7
Demand			30	40	50

- Opening a warehouse creates a fixed cost.
- Shipping cost depends on warehouse and customer.
- Shipments are allowed only from open warehouses.

Facility opening: variables and objective

Binary opening variables:

$$y_i = \begin{cases} 1, & \text{if warehouse } i \text{ is opened,} \\ 0, & \text{otherwise,} \end{cases} \quad i = 1, 2.$$

Continuous shipment variables:

$q_{ij} \geq 0$ = units shipped from warehouse i to customer j .

$$\begin{aligned} \min \quad & 400y_1 + 500y_2 \\ & + 4q_{11} + 6q_{12} + 9q_{13} + 5q_{21} + 4q_{22} + 7q_{23}. \end{aligned}$$

Facility opening: Big-M constraints

Demand constraints:

$$q_{11} + q_{21} = 30, \quad q_{12} + q_{22} = 40, \quad q_{13} + q_{23} = 50.$$

Big-M capacity constraints:

$$q_{11} + q_{12} + q_{13} \leq 80y_1,$$

$$q_{21} + q_{22} + q_{23} \leq 100y_2.$$

Variable restrictions:

$$y_i \in \{0, 1\}, \quad i = 1, 2, \quad q_{ij} \geq 0.$$

Why this is Big-M

Here M is the warehouse capacity. If $y_i = 0$, all shipments from warehouse i must be zero.

Summary

- 1 Binary variables represent yes-or-no decisions.
- 2 Assignment models use exactly-one constraints.
- 3 LP relaxations give useful bounds, but may produce fractional solutions.
- 4 Branch-and-Bound branches on fractional variables and prunes nodes using bounds.
- 5 Logical constraints can often be written as simple linear inequalities.
- 6 Big-M constraints link continuous decisions to binary on/off decisions.

Facility opening solution

Open both warehouses. One optimal shipment plan is: $q_{11} = 30$, $q_{22} = 40$, $q_{23} = 50$, with total cost 1530.

Additional practice 1: 0-1 knapsack

A student wants to pack items into a backpack with capacity 12.

Item	1	2	3	4	5
Value	20	30	35	12	3
Weight	2	5	7	3	1

Formulate an IP that chooses a subset of items to maximize total value without exceeding capacity.

Practice 1 solution: 0-1 knapsack

Step 1: define variables.

$$x_i = \begin{cases} 1, & \text{if item } i \text{ is selected,} \\ 0, & \text{otherwise.} \end{cases}$$

Step 2: maximize total value.

$$\max \quad 20x_1 + 30x_2 + 35x_3 + 12x_4 + 3x_5.$$

Step 3: enforce backpack capacity.

$$2x_1 + 5x_2 + 7x_3 + 3x_4 + x_5 \leq 12.$$

$$x_i \in \{0, 1\}, \quad i = 1, \dots, 5.$$

Additional practice 2: set covering

A city wants to open service centers to cover five districts.

Center	Cost	Districts covered
A	8	1, 2, 4
B	6	2, 3
C	7	3, 4, 5
D	5	1, 5

Formulate an IP that minimizes total opening cost while covering every district at least once.

Practice 2 solution: set covering

Step 1: define $x_A, x_B, x_C, x_D \in \{0, 1\}$, where $x_k = 1$ if center k is opened.

Step 2: minimize total cost.

$$\min \quad 8x_A + 6x_B + 7x_C + 5x_D.$$

Step 3: cover each district at least once.

$x_A + x_D \geq 1$	district 1,
$x_A + x_B \geq 1$	district 2,
$x_B + x_C \geq 1$	district 3,
$x_A + x_C \geq 1$	district 4,
$x_C + x_D \geq 1$	district 5.

$$x_A, x_B, x_C, x_D \in \{0, 1\}.$$

Additional practice 3: set packing

A company can select projects, but some pairs cannot be selected together.

Project	1	2	3	4	5
Profit	16	22	18	24	12

Conflicting pairs:

$$(1, 3), \quad (1, 4), \quad (2, 4), \quad (3, 5).$$

Formulate an IP that maximizes total profit subject to these incompatibility constraints.

Practice 3 solution: set packing

Step 1: define $x_i \in \{0, 1\}$, where $x_i = 1$ if project i is selected.

Step 2: maximize total profit.

$$\max \quad 16x_1 + 22x_2 + 18x_3 + 24x_4 + 12x_5.$$

Step 3: add one constraint for each conflicting pair.

$$x_1 + x_3 \leq 1, \quad x_1 + x_4 \leq 1,$$

$$x_2 + x_4 \leq 1, \quad x_3 + x_5 \leq 1.$$

$$x_i \in \{0, 1\}, \quad i = 1, \dots, 5.$$

Additional practice 4: shortest path

A student wants to travel from Home H to Campus C . The available directed links and travel times are:

Link	$H \rightarrow A$	$H \rightarrow B$	$A \rightarrow B$	$A \rightarrow C$	$B \rightarrow A$	$B \rightarrow C$
Time	4	2	1	7	3	5

Formulate a binary IP that selects a shortest route from H to C .

Practice 4 solution: shortest path

Step 1: define $x_{ij} \in \{0, 1\}$, where $x_{ij} = 1$ if link $i \rightarrow j$ is used.

Step 2: minimize total travel time.

$$\min \quad 4x_{HA} + 2x_{HB} + x_{AB} + 7x_{AC} + 3x_{BA} + 5x_{BC}.$$

Step 3: use flow-balance constraints.

$$x_{HA} + x_{HB} = 1$$

leave Home,

$$x_{AC} + x_{BC} = 1$$

enter Campus,

$$x_{HA} + x_{BA} = x_{AB} + x_{AC}$$

node A balance,

$$x_{HB} + x_{AB} = x_{BA} + x_{BC}$$

node B balance.

$$x_{HA}, x_{HB}, x_{AB}, x_{AC}, x_{BA}, x_{BC} \in \{0, 1\}.$$

Additional practice 5: prerequisite selection

A company can select optional software modules under a budget of 13.

Module	1	2	3	4	5
Benefit	12	18	20	15	8
Cost	4	5	6	4	2

Additional rules:

- Module 2 can be selected only if Module 1 is selected.
- Module 4 can be selected only if Module 3 is selected.
- Modules 3 and 5 cannot both be selected.

Formulate an IP that maximizes total benefit.

Practice 5 solution: prerequisite selection

Step 1: define $x_i \in \{0, 1\}$, where $x_i = 1$ if module i is selected.

Step 2: maximize total benefit.

$$\max \quad 12x_1 + 18x_2 + 20x_3 + 15x_4 + 8x_5.$$

Step 3: add budget and logical constraints.

$$4x_1 + 5x_2 + 6x_3 + 4x_4 + 2x_5 \leq 13.$$

$$x_2 \leq x_1, \quad x_4 \leq x_3, \quad x_3 + x_5 \leq 1.$$

$$x_i \in \{0, 1\}, \quad i = 1, \dots, 5.$$

Key trick

“Option 2 only if option 1” becomes $x_2 \leq x_1$.

Additional practice 6: movie night selection

A group wants to choose movies for a movie night. The total available time is 12 half-hours.

Movie	1	2	3	4	5	6
Score	9	7	8	10	6	7
Length	4	3	4	5	2	3

Rules: at least one comedy must be chosen: movies 2 or 5. Movie 4 is a sequel, so it can be chosen only if movie 1 is chosen. Movies 3 and 6 should not both be chosen.

Formulate an IP that maximizes total score.

Practice 6 solution: movie night selection

Step 1: define $x_i \in \{0, 1\}$, where $x_i = 1$ if movie i is chosen.

Step 2: maximize total score.

$$\max \quad 9x_1 + 7x_2 + 8x_3 + 10x_4 + 6x_5 + 7x_6.$$

Step 3: add time and logical constraints.

$$4x_1 + 3x_2 + 4x_3 + 5x_4 + 2x_5 + 3x_6 \leq 12.$$

$$x_2 + x_5 \geq 1, \quad x_4 \leq x_1, \quad x_3 + x_6 \leq 1.$$

$$x_i \in \{0, 1\}, \quad i = 1, \dots, 6.$$

Additional practice 7: cafe shift scheduling

A cafe chooses shifts to cover four busy periods.

Shift	Cost	Morning	Lunch	Afternoon	Evening
1	50	1	0	0	0
2	65	1	1	0	0
3	60	0	1	1	0
4	55	0	0	1	1
5	45	0	0	0	1

Required coverage: Morning 1, Lunch 2, Afternoon 2, Evening 1.

Formulate an IP that minimizes total shift cost.

Practice 7 solution: cafe shift scheduling

Step 1: define $x_s \in \{0, 1\}$, where $x_s = 1$ if shift s is selected.

Step 2: minimize total cost.

$$\min \quad 50x_1 + 65x_2 + 60x_3 + 55x_4 + 45x_5.$$

Step 3: cover each busy period.

$x_1 + x_2 \geq 1$	Morning,
$x_2 + x_3 \geq 2$	Lunch,
$x_3 + x_4 \geq 2$	Afternoon,
$x_4 + x_5 \geq 1$	Evening.

$$x_s \in \{0, 1\}, \quad s = 1, \dots, 5.$$

Additional practice 8: exam timetabling

Four exams must be assigned to one of three time slots. Some pairs have common students and cannot be scheduled at the same time.

Conflicting pairs:

$$(1, 2), \quad (1, 3), \quad (2, 4), \quad (3, 4).$$

Using slot 3 is less preferred, with penalties:

$$p_1 = 3, \quad p_2 = 4, \quad p_3 = 2, \quad p_4 = 5.$$

Formulate an IP that assigns each exam to exactly one slot and minimizes total slot-3 penalty.

Practice 8 solution: exam timetabling

Step 1: define $x_{it} \in \{0, 1\}$, where $x_{it} = 1$ if exam i is assigned to slot t .

Step 2: minimize total penalty for slot 3.

$$\min \quad 3x_{13} + 4x_{23} + 2x_{33} + 5x_{43}.$$

Step 3: assign each exam exactly once.

$$x_{i1} + x_{i2} + x_{i3} = 1, \quad i = 1, \dots, 4.$$

Step 4: prevent conflicting exams from sharing a slot.

$$x_{it} + x_{jt} \leq 1, \quad (i, j) \in \{(1, 2), (1, 3), (2, 4), (3, 4)\}, \quad t = 1, 2, 3.$$

Additional practice 9: fantasy lineup

Select a fantasy team under a salary cap of 30.

Player	1	2	3	4	5	6	7
Position	GK	DEF	DEF	DEF	MID	MID	MID
Salary	7	6	5	4	8	7	6
Points	10	9	8	6	12	11	9

Choose exactly 1 goalkeeper, 2 defenders, and 2 midfielders.
Formulate an IP that maximizes total projected points.

Practice 9 solution: fantasy lineup

Step 1: define $x_i \in \{0, 1\}$, where $x_i = 1$ if player i is selected.

Step 2: maximize total points.

$$\max \quad 10x_1 + 9x_2 + 8x_3 + 6x_4 + 12x_5 + 11x_6 + 9x_7.$$

Step 3: add salary and position constraints.

$$7x_1 + 6x_2 + 5x_3 + 4x_4 + 8x_5 + 7x_6 + 6x_7 \leq 30.$$

$$x_1 = 1, \quad x_2 + x_3 + x_4 = 2, \quad x_5 + x_6 + x_7 = 2.$$

$$x_i \in \{0, 1\}, \quad i = 1, \dots, 7.$$

Additional practice 10: food-truck coverage

A food-truck company can park at at most two locations to cover neighborhoods.

Location	Parking cost	Neighborhoods covered
A	5	1, 2, 4
B	4	2, 3
C	6	3, 4, 5
D	3	1, 5

The company wants to maximize the number of covered neighborhoods and may spend at most 9 on parking.

Formulate an IP using coverage variables.

Practice 10 solution: food-truck coverage

Step 1: define $x_k = 1$ if location k is chosen, and $y_j = 1$ if neighborhood j is covered.

Step 2: maximize covered neighborhoods.

$$\max \quad y_1 + y_2 + y_3 + y_4 + y_5.$$

Step 3: add selection, budget, and coverage-linking constraints.

$$x_A + x_B + x_C + x_D \leq 2, \quad 5x_A + 4x_B + 6x_C + 3x_D \leq 9.$$

$$y_1 \leq x_A + x_D, \quad y_2 \leq x_A + x_B, \quad y_3 \leq x_B + x_C,$$

$$y_4 \leq x_A + x_C, \quad y_5 \leq x_C + x_D.$$

$$x_k \in \{0, 1\}, \quad y_j \in \{0, 1\}.$$